Runbook: invoke-technician-onetouch.ps1

Oct 2, 2026 · @Jackie Nguyen

Overview

`invoke-technician-onetouch.ps1` turns a Windows 11 Pro device into a locked Skriin AI TV kiosk in one run: optional activation, the SkriinWifiSetup service, then Assigned Access pointing Edge at `KioskUrl`. It lives in `utilities/kiosk-deployment-package/` and is the only script a technician should run directly; it calls the others.

When it succeeds, the device boots straight into full-screen Edge on `http://127.0.0.1:5757/` (the on-screen Wi-Fi picker), with the owner admin account reachable via the breakout sequence (default Ctrl+Alt+Shift+S).

Script	Called when	Job
`invoke-technician-onetouch.ps1`	Technician runs it	Validates config, relaunches as SYSTEM, orchestrates the rest
`apply-activation.ps1`	`ProductKey` or `ActivationCsvPath` set	Installs and activates a Windows product key
`../wifi-setup/install.ps1`	`InstallWifiSetup` true (default)	Installs and starts the SkriinWifiSetup service
`setup-kiosk.ps1`	Always	Hardening, owner admin account, Assigned Access, optional reboot
`reset-kiosk.ps1`	Manually, for recovery	Clears Assigned Access and reverts selected hardening

Prerequisites

Run it from an elevated PowerShell on the device's **local console**, with the whole `utilities` folder copied over so `kiosk-deployment-package` and `wifi-setup` sit side by side.

- [ ] Windows 11 Pro with Microsoft Edge Stable installed (setup fails if `msedge.exe` is missing).
- [ ] Elevated PowerShell (`#Requires -RunAsAdministrator`).
- [ ] Local console session. An `RDP-*` session is refused unless you pass `-AllowRdpSession`.
- [ ] `technician-config.json` beside the script (copy from `technician-config.sample.json`), or a path passed via `-ConfigPath`.
- [ ] `wifi-setup\WifiSetupService\publish\win-x64\WifiSetupService.exe` exists. Build it with `wifi-setup\build.ps1` on a machine with the .NET 8 SDK.
- [ ] Network access if online activation is wanted.

Invocation

```
cd <utilities>\kiosk-deployment-package
.\invoke-technician-onetouch.ps1                       # uses .\technician-config.json
.\invoke-technician-onetouch.ps1 -ConfigPath D:\client-a.json
.\invoke-technician-onetouch.ps1 -AllowRdpSession     # lab use only
```

Config keys

Key	Required	Default	Effect
`KioskUrl`	Yes	none	URL Edge opens in kiosk mode; normally `http://127.0.0.1:5757/`
`OwnerAdminPasswordPlainText`	Yes	none	Password for the owner admin account
`OwnerAdminUserName`	No	`skriinadmin`	Local admin account created or updated
`OwnerAdminFullName`	No	`Skriin Admin`	Display name of that account
`OwnerAdminLogonWorkflow`	No	`ShowAdminTile`	`ShowAdminTile` or `RequireTypedCredentials`; anything else fails
`DisplayName`	No	`Skriin AI TV`	Assigned Access profile and kiosk account name
`BreakoutSequence`	No	`Ctrl+Alt+Shift+S`	Key combo that exits the kiosk to sign-in

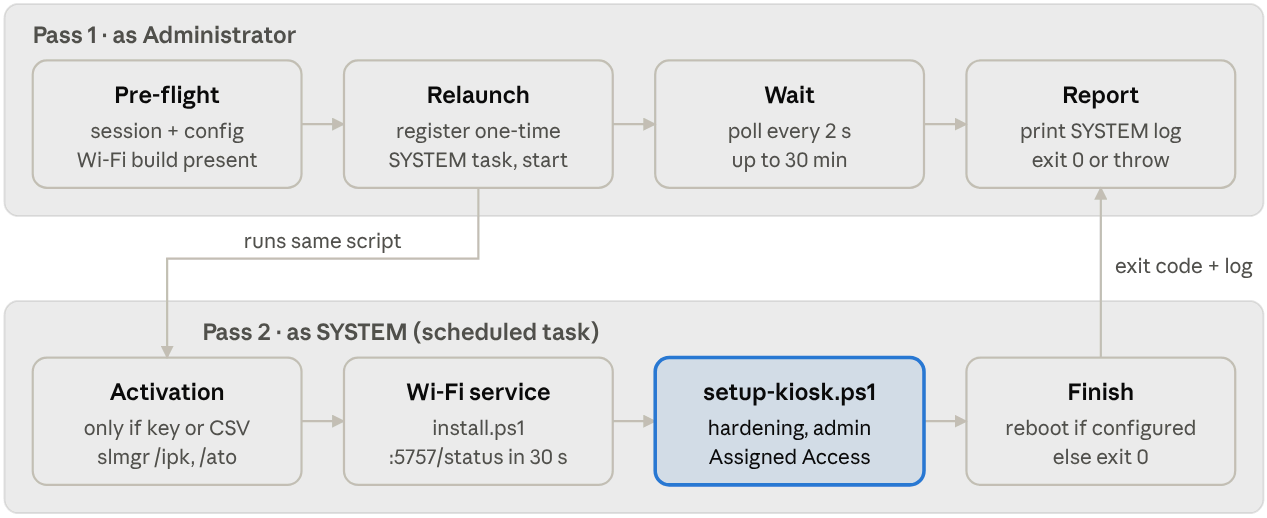
Key	Required	Default	Effect
`CameraSiteUrl`	No	`https://skriin.com/`	Site granted camera/mic in Edge policy
`InstallWifiSetup`	No	`true`	Install/update SkriinWifiSetup during the run
`WifiSetupPath`	No	`..\wifi-setup`	Where the Wi-Fi installer lives
`LogonBackgroundImagePath`	No	none	.jpg/.png used as lock/logon background
`ConfigureScancodeHardening`	No	off	Disables Win and Menu keys
`RebootAfterApply`	No	off	Reboots 5 s after a successful apply
`ProductKey`	No	none	Key to install with slmgr
`ActivationCsvPath`	No	none	CSV mapping BIOS serial to product key
`ActivationCsvSerialColumn` / `ActivationCsvKeyColumn`	No	`SerialNumber` / `ProductKey`	CSV column names

Relative paths in the config resolve against the `kiosk-deployment-package` folder. The config holds a plaintext password, so delete or remove it from the device after the run.

Execution flow

The script runs twice: once as you (Administrator) to validate and hand off, then again as SYSTEM to do the real work. Any `throw` stops the run, because `$ErrorActionPreference` is `Stop`.

Pass 1 validates and hands off; Pass 2 as SYSTEM does the work



one-touch flow · 2 passes, 8 stages

Pass 1 never changes the device beyond a temporary scheduled task; all configuration happens in Pass 2, and its exit code decides what Pass 1 reports.

Pass 1: as Administrator (pre-flight and relaunch)

1. **Session check.** If `SESSIONNAME` starts with `RDP-` and `-AllowRdpSession` is absent, stop.
2. **Load config.** Use `-ConfigPath` or default `technician-config.json`; stop if missing. Parse JSON.
3. **Validate.** Stop if `KioskUrl` or `OwnerAdminPasswordPlainText` is empty. If `ActivationCsvPath` is set, the CSV must exist.
4. **Wi-Fi pre-check.**
   - `InstallWifiSetup` true: resolve `WifiSetupPath`; require `install.ps1` and the published `WifiSetupService.exe`.
   - `InstallWifiSetup` false: if `KioskUrl` is loopback and the SkriinWifiSetup service is not installed, stop (the kiosk would boot to a blank page).
5. **Relaunch as SYSTEM.** Assigned Access can only be written by LocalSystem, so the script:
   - registers a one-time scheduled task `SkriinKioskOneTouch-<8 hex>` running `powershell.exe -File <this script> -ConfigPath <config>` as SYSTEM, 30-minute limit;

- starts it and polls every 2 s until it stops running or 30 minutes pass;
- reads `LastTaskResult`, then unregisters the task (always, in `finally`);
- prints the new `logs\technician-run-*.log` the SYSTEM run wrote;
- stops with an error if the exit code is non-zero, otherwise prints "completed successfully (ran as SYSTEM)".

Pass 2: as SYSTEM (the actual work)

Steps 1–4 repeat, then the identity check passes and the script starts a transcript at `logs\technician-run-<yyyyMMdd-HHmmss>.log`.

6. **Activation** (only if `ProductKey` or `ActivationCsvPath` is set) via `apply-activation.ps1`. Key source, first match wins: direct `ProductKey` → CSV row matching the BIOS serial → firmware-embedded OA3 key (only when no CSV was given). Runs `slmgr /ipk <key>` then `slmgr /ato`, and prints `ActivationStatus=Licensed`, `UnknownOrUnlicensed` or `Skipped`. None of these stop the run.
7. **Wi-Fi setup service** (if `InstallWifiSetup`) via `wifi-setup\install.ps1`:
   - creates `C:\Program Files\Skriin\WifiSetup` and `C:\ProgramData\Skriin`;
   - stops and deletes any existing `SkriinWifiSetup` service, then copies the published build in;
   - forces Location Services on (policy, consent store, `lfsvc` master switch) because `netsh wlan` needs it;
   - registers the service (Automatic start), grants SYSTEM full control on the data dir, starts it.

   The one-touch script then requires the service to be `Running` and `http://127.0.0.1:5757/status` to answer within 30 s.
8. **Kiosk configuration** via `setup-kiosk.ps1`, with defaults filled in for unset keys:
   a. Refuses to run unless SYSTEM; finds `msedge.exe`.
   b. Power: monitor, sleep and hibernate timeouts to 0; hibernate off; lock screen disabled.
   c. Logon background (if set): copied to `C:\KioskSetup\Assets`, set via Personalization policy and CSP keys.
   d. Screen saver off for the current hive and `.DEFAULT`.
   e. Edge policies: no first-run, sign-in, sync, promos, password manager or relaunch prompts.

   f. Camera/mic allow-lists rebuilt for `CameraSiteUrl` (and `*.` subdomains) plus the `KioskUrl` origin.
   g. Toasts and Notification Center off; Fast User Switching left on; Task Manager disabled.
   h. Windows Update: no auto-reboot with users signed in, install at 03:00, active hours 06:00–23:00.
   i. Quiet boot (`bcdedit bootux disabled`, `quietboot yes`), best effort.
   j. Owner admin account created or its password reset, and added to Administrators.
   k. Scancode hardening (if set): Left/Right Win and Menu keys disabled.
   l. Account-picker policy backed up once to `C:\KioskSetup\account-picker-policy-backup.json`; logon workflow policy applied.
   m. **Assigned Access**: writes `C:\KioskSetup\assigned-access.xml` (Edge `--kiosk <KioskUrl> --edge-kiosk-type=fullscreen`, breakout key, auto-logon account named `DisplayName`), checks it parses, and sets it on `MDM_AssignedAccess`. On failure it dumps Assigned Access event logs, clears the existing config, waits 10 s and retries once.
   n. Finds the new kiosk account and hides every other local account from the sign-in screen, except the owner admin when `ShowAdminTile` is used.
   o. Reboots after 5 s if `RebootAfterApply`; otherwise recommends a reboot.
9. **Finish.** Prints "One-touch workflow completed successfully" and the log path, stops the transcript, exits 0. Pass 1 picks this up and reports success.

Outputs and verification

The run log and `C:\KioskSetup\assigned-access.xml` are the two files to check first.

Path	Written by	Contents
`kiosk-deployment-package\logs\technician-run-<stamp>.log`	One-touch (SYSTEM pass)	Full transcript of activation, Wi-Fi install and kiosk setup
`C:\KioskSetup\assigned-access.xml`	setup-kiosk	The Assigned Access config that was applied

Path	Written by	Contents
`C:\KioskSetup\account-picker-policy-backup.json`	setup-kiosk	Sign-in screen policy before the first run (used by `reset-kiosk.ps1 -RestoreAccountPickerPolicy`)
`C:\KioskSetup\assigned-access-kiosk-user.txt`	setup-kiosk	Name of the inferred kiosk auto-logon account
`C:\KioskSetup\Assets\logon-background.*`	setup-kiosk	Staged lock/logon image
`C:\Program Files\Skriin\WifiSetup\`	install.ps1	SkriinWifiSetup binaries
`C:\ProgramData\Skriin\`	install.ps1	Service data dir (SYSTEM full control)

Verify after the run

- [ ] Console ends with "One-touch workflow completed successfully (ran as SYSTEM)." (not shown if the device rebooted first; check the log instead).
- [ ] Log contains `[OK] Wi-Fi setup service is running and responding` and `Assigned Access kiosk configuration applied successfully.`
- [ ] If activation ran: log shows `ActivationStatus=Licensed`.
- [ ] `Get-Service SkriinWifiSetup` shows Running, Automatic.
- [ ] After reboot: device auto-signs in and Edge opens full screen on the Wi-Fi picker.
- [ ] Breakout sequence returns to sign-in; owner admin can sign in.
- [ ] Webcam works in the picker's QR scan and on the Skriin site.

Troubleshooting

Every failure before step 8.13 is safe to fix and re-run; the script is idempotent. A failure at Assigned Access can leave the device with **no** kiosk config, so re-run once the cause is fixed.

Error text (starts with)	Where	Cause	Fix
`Assigned Access configuration must run from a local console session`	Pass 1	Running over RDP	Use the local console, or `-AllowRdpSession` in a lab

Error text (starts with)	Where	Cause	Fix
`ConfigPath was not specified and default config file was not found` / `Config file not found`	Pass 1	No config	Create `technician-config.json` from the sample
`Config must include KioskUrl` / `OwnerAdminPasswordPlainText`	Pass 1	Required key empty	Fill it in
`ActivationCsvPath was specified, but the file was not found`	Pass 1	Wrong CSV path	Fix path (relative to the package folder)
`Wi-Fi setup folder not found` / `installer not found`	Pass 1	`wifi-setup` not copied beside the package	Copy the whole `utilities` folder, or set `WifiSetupPath`
`Wi-Fi setup service has not been built`	Pass 1	No published exe	Run `wifi-setup\build.ps1` on a .NET 8 SDK machine, copy over
`KioskUrl points at the local Wi-Fi setup service ... not installed`	Pass 1	`InstallWifiSetup: false` but service absent	Set it to `true`
`SYSTEM run of the one-touch workflow failed (exit code N)`	Pass 1	Pass 2 threw	Read the SYSTEM log printed above it. Exit code `267009` means it was still running at the 30-minute limit
`SkriinWifiSetup service is not running after install`	Step 7	Service failed to start	Check Event Viewer > Application; try `Start-Service SkriinWifiSetup`
`... did not respond within 30 seconds`	Step 7	Port 5757 not answering	Check the service and that nothing else holds port 5757
`Microsoft Edge was not found`	Step 8.1	Edge missing	Install Edge Stable
`OwnerAdminLogonWorkflow must be ...`	Step 8	Typo in config	Use `ShowAdminTile` or `RequireTypedCredentials`
`Logon background image must be a .jpg, .jpeg, or .png`	Step 8.3	Wrong file type	Use a supported image
`Applying Assigned Access failed twice`	Step 8.13	Windows rejected the XML	Read the diagnostics block in the log and `C:\KioskSetup\assigned-access.xml`, fix, re-run

Behaviours worth knowing

- **Reboot cuts the relay.** With `RebootAfterApply`, the device restarts inside the SYSTEM task, so Pass 1 never prints the success line or the log. Check `logs\` after reboot.
- **Activation never fails the run.** slmgr errors and `UnknownOrUnlicensed` are only reported. Search the log for `ActivationStatus`.
- **CSV with no match skips activation.** When `ActivationCsvPath` is set, the firmware key is not tried as a fallback.
- **Re-runs reset the owner admin password** to the config value and reinstall the Wi-Fi service.
- **Screen-saver HKCU keys** land in SYSTEM's hive during Pass 2, not the kiosk user's; the `.DEFAULT` keys are what apply.
- **Kiosk account hiding is skipped** if the kiosk account can't be inferred (warning in log); the kiosk still works.

Roll back

```
.\reset-kiosk.ps1 -RestoreAccountPickerPolicy -EnableTaskManager -
RemoveScancodeHardening -RemoveLogonBackgroundPolicy -RebootAfterReset
```

This clears Assigned Access and reverts the selected hardening. It does not uninstall SkriinWifiSetup.